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LEE KONG CHIAN SCHOOL OF MEDICINE

Modelling the interplay between doxycycline post-exposure prophylaxis, vaccination, and antimicrobial resistance in *Neisseria gonorrhoeae*

Zihao Wang^{1,7}, Dariya Nikitin², Liang En Wee⁴, Martin T.W. Chio⁵, Rayner Kay Jin Tan³, Yi Wang³, David N. Fisman⁶, Lilith K. Whittles², Jue Tao Lim^{1,7,*}

¹ Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore
³ Saw Swee Hock School of Public Health, National University of Singapore, Singapore
⁵ National Skin Centre, Singapore
⁷ National Centre for Infectious Diseases, Singapore

² Department of Infectious Disease Epidemiology, Imperial College London, London, United Kingdom
⁴ Singapore General Hospital, Singapore
⁶ Dalla Lana School of Public Health, University of Toronto, Toronto, Canada
* Corresponding author. E-mail: jue tao.lim@ntu.edu.sg

Motivation

- Doxycycline, a tetracycline-class antibiotic, has recently gained renewed interest through its use as post-exposure prophylaxis (doxy-PEP) for preventing syphilis.
- However, its large-scale deployment has raised concerns regarding potential impacts on antimicrobial resistance (AMR) in *Neisseria gonorrhoeae*.

Objective

- Investigate how doxy-PEP for syphilis prevention may exacerbate the spread of strain-resistant *N. gonorrhoeae* among men who have sex with men (MSM).
- Evaluate how a moderate-protection vaccination could mitigate gonorrhoea AMR emergence and offset the evolutionary risks introduced by doxy-PEP for syphilis.

Strain-Specific Gonorrhea Transmission Model

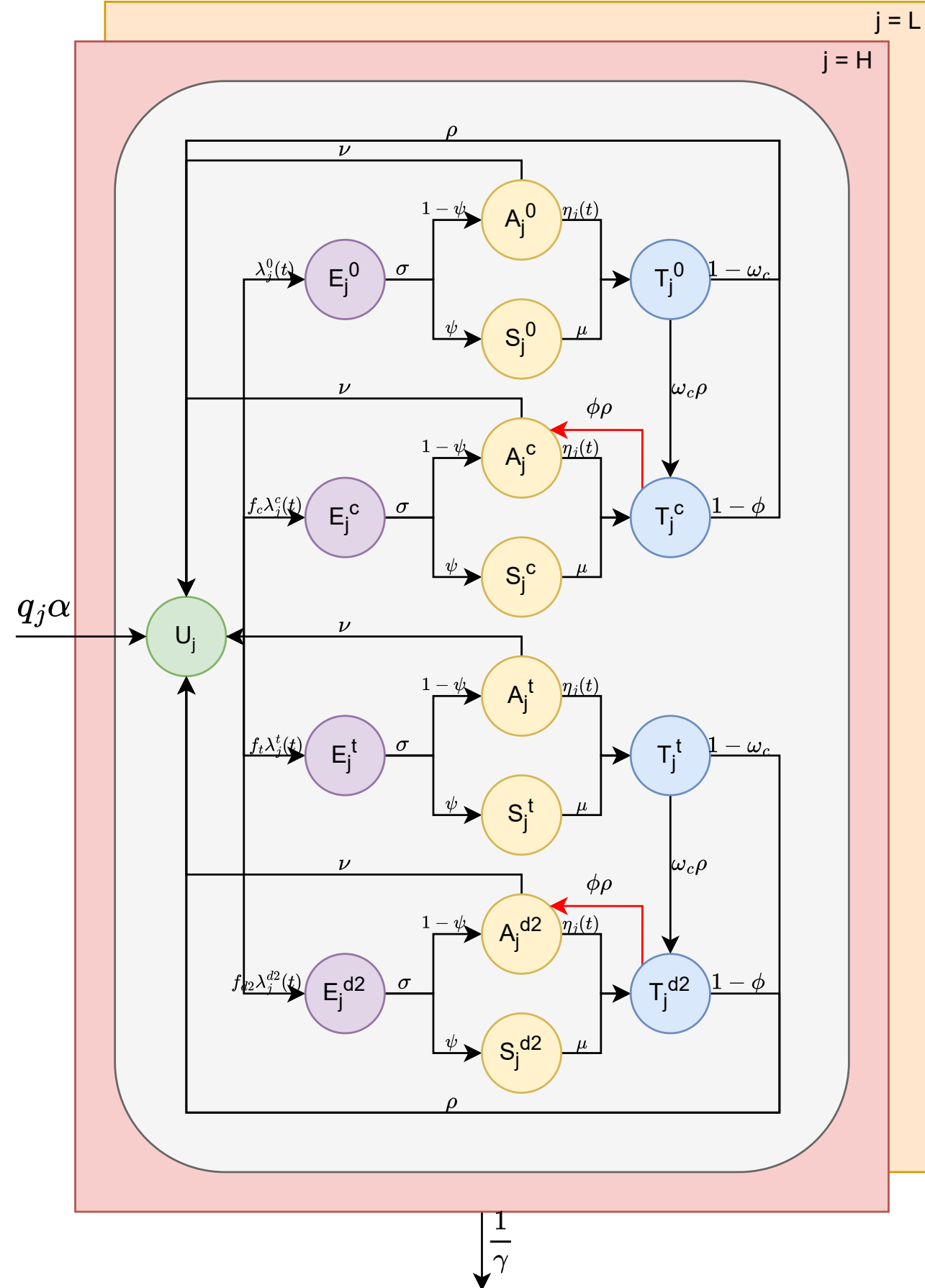


Fig. 1: Model architecture for gonorrhea transmission.

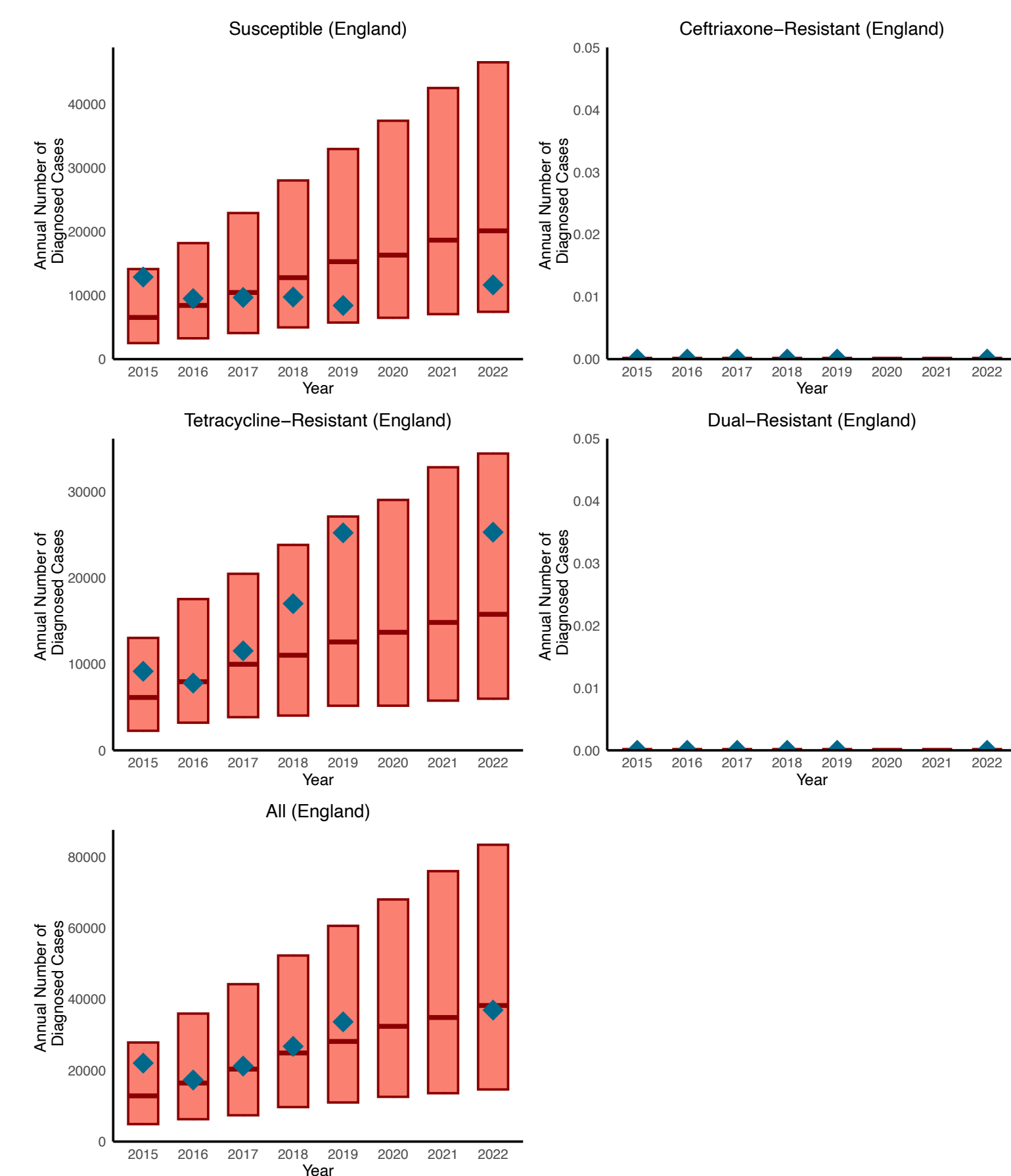


Fig. 2: Simulated epidemic trajectories.

Two main types of data for model calibration

1. Epidemiological data on MSM syphilis incidence in England:

- Annual numbers of gonorrhoea diagnoses, tests, percentage ceftriaxone-resistant, and percentage tetracycline-resistant among MSM from 2015 to 2022 were retrieved from the Genitourinary Medicine Clinic Activity Dataset.
- Annual numbers of MSM gonorrhoea symptomatic and asymptomatic diagnoses, and sample size from 2015 to 2022 were retrieved from Gonococcal Resistance to Antimicrobials Surveillance Program.

2. Sexual behavioural data from England.

The Evolutionary Frontier

- We focus on four phenotypes defining the current clinical landscape:
- Susceptible (wild-type)
- Tet-R (Tetracycline-resistant MIC>1mg/L)
- Cef-R (Ceftriaxone-resistant MIC>0.125mg/L)
- Dual-R

Doxy-PEP with Vaccination Model for Gonorrhea

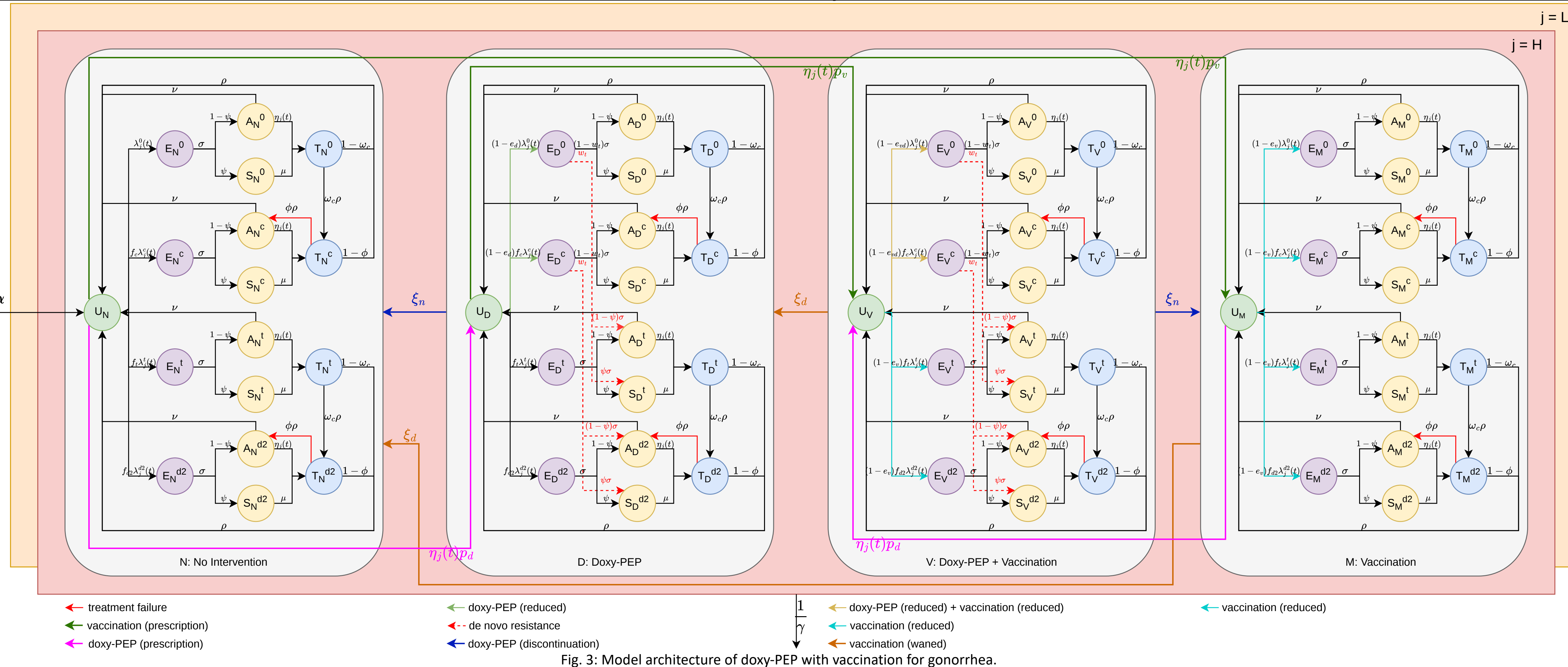


Fig. 3: Model architecture of doxy-PEP with vaccination for gonorrhea.

The “Selection Collision” Model

- Our model simulates the intersection of two distinct evolutionary pressures:
- Doxy-PEP Pressure:** Elevates Tet-R from a common bystander to a dominant phenotype (the “evolutionary cost” of syphilis prevention).
- Treatment Pressure:** Reliance on ceftriaxone as the last-line monotherapy drives Cef-R emergence (rising in England since 2021).
- The Nightmare Scenario:
- The emergence of a **Dual-R “Superbug”**: Simultaneously unpreventable by doxy-PEP and potentially untreatable by standard-of-care ceftriaxone.

Simulation Results

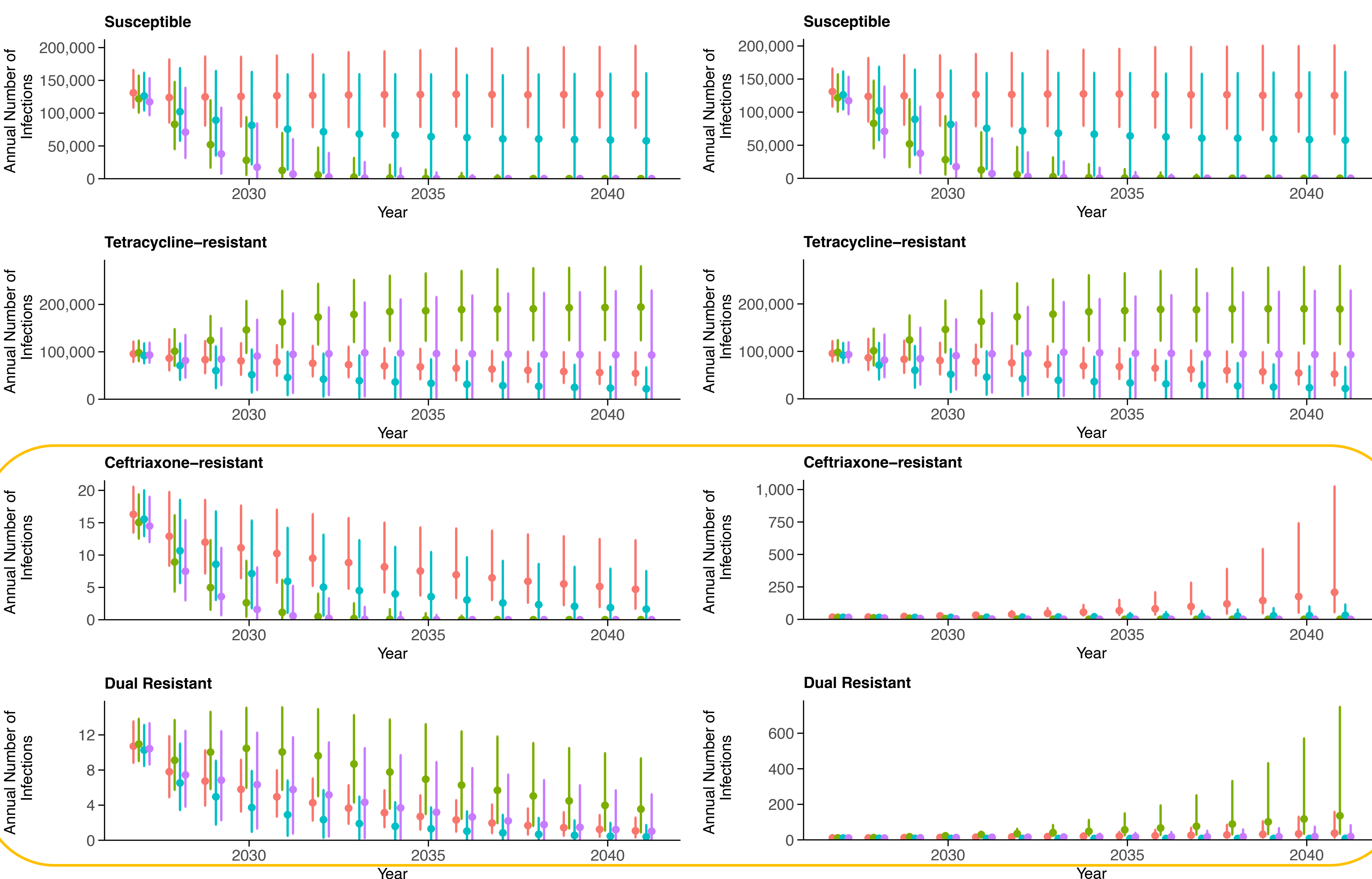


Fig. 4: Left panel: Annual number of gonorrhoea infections (near-zero ceftriaxone treatment failure rate). Right panel: Annual number of gonorrhoea infections (20% ceftriaxone treatment failure rate).

Recommendation

Doxy-PEP may “win” the immediate war against syphilis, but a multi-faceted approach - integrating non-selective tools like moderate-protection vaccination - is essential to avoid “losing” the long-term fight against multi-drug resistant gonorrhoea.